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Pyrolysis of $C_6D_5CH_3$: Rate constants and branching ratios in the high-temperature thermal decomposition of toluene

R. Sivaramakrishnan, J.V. Michael *

Chemical Sciences & Engineering Division, Argonne National Laboratory, Argonne, IL 60439, USA

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Abstract

The thermal decomposition of toluene-d_5 ($C_6D_5CH_3$) has been studied at high temperatures with the reflected shock-tube technique using H- and D-atom ARAS. The experiments were performed at high-T (1469–1859 K) at nominal pressures ≈0.25–1.50 atm.

The present study utilizing the ultra-sensitive H-atom ARAS technique has provided a direct measurement of the branching ratio in the two-channel high-temperature thermal decomposition of toluene-d_5 giving (1) $C_6D_5CH_2 + H$ and (2) $C_6D_5 + CH_3$. Fall-off is observed in both decomposition channels at high-T (>1700 K), but the lower-T rate constants can be well represented by simple Arrhenius expressions in units of s^{-1} as,

$$k_1 = 6.91 \times 10^{13}\ \exp(-40180/T) \quad (1469–1714\ \mathrm{K})$$

$$k_2 = 5.99 \times 10^{15}\ \exp(-50060/T) \quad (1469–1714\ \mathrm{K})$$

The experimental results were also used to obtain rate constants for total toluene-d_5 decomposition. Arrhenius analysis gives in units of s^{-1},

$$k_{\mathrm{Total}} = 2.21 \times 10^{14}\ \exp(-41760/T) \quad (1469–1714\ \mathrm{K})$$

The isotope effect is minimal and the present results therefore represent direct measurements of branching ratios and rate constants for toluene decomposition. The branching ratios to benzyl + H vary from ≈0.9 at lower-T (<1600 K) to ≈0.75 at higher-T (>1700 K). The excellent agreement between the present experiments and the theoretical predictions by Klippenstein et al. lead us to conclude that the high-T thermal decomposition of toluene is now well-characterized.
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Keywords: Shock tube; Aromatics; Soot precursor; Atomic resonance absorption spectroscopy (ARAS); Gasoline and diesel surrogate

1. Introduction

Toluene, the simplest alkyl-substituted aromatic species, is a major component in gasoline and diesel (5–25% by volume) and represents a model aromatic

* Corresponding author. Address: Chemical Sciences & Engineering Division, Argonne National Laboratory, Bldg. 200, Rm. D-193, Argonne, IL 60439, USA. Fax: +1 630 252 9570.
E-mail address: jmichael@anl.gov (J.V. Michael).

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surrogate species for these fuels. Despite extensive experimental and modeling studies on toluene combustion, its low- and high-temperature mechanisms are poorly understood [1] largely due to a lack of consensus on the thermal dissociation rate constants. The thermal dissociation at high temperatures is a two-channel process, one involving a C–H split giving H + benzyl, and the other a C–C split giving methyl + phenyl. This two-channel dissociation is complicated by fast recombinations and subsequent bimolecular reactions, particularly at high-T. Interestingly, the dissociation of the major product, the benzyl radical, is faster than toluene dissociation by a factor of 2–4 at high-T, >1500 K [2], thereby making chemical isolation of the primary dissociation channels difficult.

Numerous reports exist on toluene pyrolysis dating back to the late 1930s. Szwarc's early experimental studies confirmed that the initiation step in toluene was indeed unimolecular C–H bond fission and also established its subsequent use as a radical scavenger [3,4]. Since this work there have been numerous experimental studies using a variety of techniques with the study by Smith [5] summarizing much of the low-temperature flow reactor studies. It is clear that at low-T, <1200 K, the pyrolysis is dominated by C–H bond fission with the high-pressure limiting rate constant set by Price's [6] flow reactor experiments.

At higher-T, >1500 K, there is no doubt that a contribution from the C–C bond fission channel exists. Shock-tube experiments [7–19] have been performed over the years using a variety of diagnostics to assess the contribution of this higher-energy dissociation channel. A number of laser excitation studies [20] have also been performed to obtain energy specific rate constants for toluene dissociation. Two recent shock-wave studies, the UV-absorption experiments by Oehlschlaeger et al. [19] and the H-ARAS experiment by Eng et al. [17], report rate constants in reasonable agreement with each other at similar pressures, ≈1.5–2 atm. A recent theoretical study by Klippenstein et al. [21] has further contributed to understanding this complicated multi-channel dissociation. However, this theoretical study [21] predicts rate constants more than a factor of two lower than those from Oehlschlaeger et al. [19] and 40–50% lower than Eng et al. [17] at $T \approx 1600$–1700 K and $P \approx 1.5$ atm. The theoretical predictions increasingly deviate from experiment at even higher-T. This disagreement may be due to secondary chemistry because of large initial concentrations of the fuel used in these studies. In particular, three secondary reactions, the recombination reaction between H and benzyl, the thermal decomposition of benzyl, and the reaction of H with toluene may significantly affect the conclusions from these recent studies.

The present experiments are a continuation of our recent studies on the thermal decomposition of benzyl radicals [22]. This benzyl decomposition study [22] has shown that the major channel at $T > 1600$ K is ring H-atom elimination. Consequently, a primary goal of the present study is to utilize the ultra-sensitive H(D)-atom atomic resonance absorption spectrometry (ARAS) technique developed in this laboratory [23,24] to determine absolute H- and/or D-atom yields. H-atom formation from $C_6D_5CH_3$ will reflect only contributions from the C–H dissociation channel in $C_6D_5CH_3$ with subsequent fast dissociation of $C_6D_5CH_2$ yielding only minor quantities of additional H-atoms [22]. D-atoms can also be measured from the major C–D dissociation channel in $C_6D_5CH_2$. Under the ultra-dilute conditions of the present experiments, reactions of H with benzyl and toluene have no effect. Therefore, the present H-atom experiments on $C_6D_5CH_3$ are close to chemical isolation, and this work, along with benzyl dissociation [22], is then a direct determination of dissociation rate constants and branching ratios in $C_6D_5CH_3$.

2. Experimental

The present experiments were performed with the reflected shock-tube technique using H(D)-atom ARAS detection. The methods and the apparatus currently being used have been previously described [23,24] and only a brief description of the experiment will be presented here.

The shock-tube was constructed entirely from a 7-m (10.2 cm o.d.) 304 stainless steel tube with the cylindrical section being separated from the He driver chamber by a 4 mil unscored 1100-H18 aluminum diaphragm. The tube was routinely pumped between experiments to less than 1.3×10^{-11} bar by an Edwards Vacuum Products Model CR100P packaged pumping system. Shock-wave velocities were measured with eight equally spaced pressure transducers (PCB Piezotronics, Inc., Model 113A21) mounted along the downstream part of the test section and recorded with a 4094C Nicolet digital oscilloscope. Temperature and density in the reflected shock-wave regime were calculated from this velocity. This procedure has been given previously, and corrections for boundary layer perturbations have been applied [25–27]. The oscilloscope was triggered by pulses derived from the last velocity gauge signal. The photometer system was radially located at 6 cm from the endplate.

For H(D)-atom detection, the lenses were crystalline MgF_2, and the resonance lamp beam intensity (filtered through 6 cm of dry air (21% O_2) to isolate the Lyman-α_H or Lyman-α_D wavelengths at 121.6 nm), was measured by an EMR G14 solar blind photomultiplier tube, as described previously, [28–31] and was recorded with a LeCroy model LC334A oscilloscope. In order to measure the fraction of non-Lyman-α_H present in the resonance absorption emission lamp, an H_2 discharge flow

system was used to create large [H] between the lamp and shock-tube lens [30], thereby removing all of the Lyman-α_H lamp emission. The H-atom experiments were then performed with the discharge system turned off. However, the D-atom experiments required metering very small amounts of D_2 into the resonance lamp such that the lamp intensity was similar to that for H-atoms. This ensures that the D-atom lamp will then be effectively unreversed; i.e., a Gaussian distribution [28]. In this case, D-atoms in the presence of H-atoms can be directly detected by carrying out the experiment with the H_2 discharge flow system turned on (i.e., removing Lyman-α_H) during the D-atom experiment.

2.1. Gases

High-purity He (99.995%), used as the driver gas, was from AGA Gases. Research grade Kr (99.999%), the diluent gas in reactant mixtures, was from Praxair, Inc. The ~10 ppm impurities (N_2 < 5 ppm, O_2 < 2 ppm, Ar < 1 ppm, CO_2 < 0.5 ppm, H_2 < 1 ppm, H_2O < 3 ppm, Xe < 2 ppm, and THC < 0.2 ppm) are all either inert or in sufficiently low concentration so as to not perturb H-atom profiles. For H(D)-atom detection, the microwave driven resonance lamp operated at 35 W and 1.4 Torr of ultra-high-purity He (99.999%) (effective Doppler temperature, 470 K, [32]). This grade of He contains a trace of hydrogenous impurities that are sufficient to give measurable Lyman-α_H radiation. $C_6D_5CH_3$ (98 at.% D) obtained from Sigma–Aldrich was further purified by bulb-to-bulb distillation, retaining only middle thirds for mixture preparation. The purity of $C_6D_5CH_3$ was confirmed by NMR to match the stated purities of the manufacturer thereby negating any possibility of scrambling in the liquid phase. Toluene has a high saturation vapor pressure, ~29 Torr at 298 K [33]. Gas mixtures were accurately prepared from pressure measurements with a Baratron capacitance manometer in an all glass high-purity vacuum line staying well below the saturation vapor pressure. The internal walls of the shock-tube are polished to a mirror finish and evacuated to $<10^{-11}$ bar before each experiment. Mixtures are filled and shocks fired within 15–20 s, thereby ensuring that there are no wall-absorption effects. Using similar techniques the thermal decomposition of C_6H_5I, a low-vapor pressure molecule with a low-saturation vapor pressure of ~1.0 Torr at 298 K, was studied in earlier work from this laboratory [34]. I-atom yields in that study [34] (unlike the Frank et al. I-ARAS study on C_6H_5I [35]) were 1.01 ± 0.1 giving us confidence in our mixture integrities and preparation techniques.

3. Results

In the present experiments, H- and D-atoms have been measured from the thermal decomposition of toluene-d_5, $C_6D_5CH_3$. A typical H-atom concentration profile from the decomposition of $C_6D_5CH_3$ at $T = 1673$ K is shown in Fig. 1. In the absence of reactions that deplete or rapidly form [H], these results will then be a direct measure of branching ratios in the two-channel thermal decomposition,

$$C_6D_5CH_3 \rightarrow C_6D_5CH_2 + H \quad (1)$$

$$C_6D_5CH_3 \rightarrow C_6D_5 + CH_3 \quad (2)$$

At the high sensitivity of the present experiment, all secondary reactions are negligible contributors with the exception being the benzyl-d_5, $C_6D_5CH_2$, thermal decomposition. The thermal decompositions of $C_6H_5CH_2$ and $C_6H_5CD_2$ have been recently studied in our laboratory [22]. The $C_6H_5CD_2$ experiments show that there are three specific pathways for benzyl destruction,

$$C_6H_5CD_2 \rightarrow C_6H_5CD + D, k_D \text{ (side-chain D-atom elimination)} \quad (3)$$

$$\rightarrow C_6H_4CD_2 + H, k_H \text{ (ring H-atom elimination)} \quad (4)$$

$$\rightarrow \text{non-atom producing products}, k_{NA} \quad (5)$$

all of which occur through multiple isomerizations. Our previous study [22] shows that over the T range, 1450–1800 K, yields of ≈0.1–0.15 D-atoms (reaction (3)) and ≈0.6–0.7 H-atoms (reaction (4)) are obtained from the D- and H-atom ARAS experiments on $C_6H_5CD_2$. Experiments on $C_6H_5CH_2$ confirmed that a total yield ≈0.7–0.8 H-atoms form (sum of reactions (3) and (4)) per molecule of benzyl that decays. Prior studies on benzyl decomposition [36] that postulated a yield of one H-atom formed per one molecule of benzyl decay were no doubt complicated by secondary reactions. The recent study from our laboratory [22] also indicated no measurable isotope effect.

The benzyl study then implies that a yield of ≈0.1–0.15 H-atoms will be produced from every molecule of $C_6D_5CH_2$ that decays. Consequently

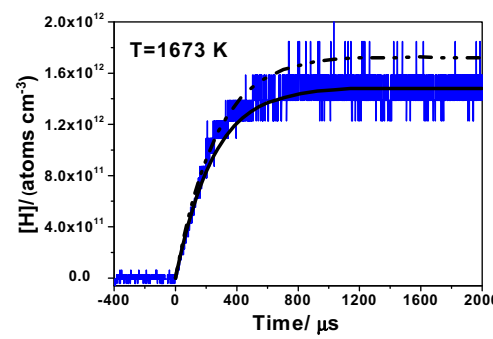

Fig. 1. [H] profile at 1673 K. Solid line – fit using the mechanism in Table S1 with $k_{total} = 4500\ s^{-1}$ and $BR_1 = 0.78$, dashed–dot–dot line – fit using the mechanism in Table S1 with $k_{total} = 4400\ s^{-1}$ and $BR_1 = 0.88$ from first-order analysis. The conditions for the experiment are $P_1 = 10.88$ Torr, $M_s = 2.597$, $\rho_5 = 2.169 \times 10^{18}$ molecules cm^{-3}, $[C_6D_5CH_3]_0 = 1.658 \times 10^{12}$ molecules cm^{-3}.

the H-atom yields from $C_6D_5CH_3$ will always include a minor contribution from $C_6D_5CH_2$ decay. This implies that first-order build-up analysis will not hold for obtaining final H-atom yields. On the other hand, first-order analysis can be used to obtain an initial first guess for the total decomposition rate and yield which can then be optimized by using a chemical kinetics model (Table S1, Supplementary data) to describe the temporal [H] profile. For H-atom experiments like Fig. 1, first-order analysis gives the simple closed form result,

$$[H]_t = \{k_1[C_6D_5CH_3]_0/k_{total}\} \times \{1 - \exp(-k_{total}t)\} \quad (6)$$

where $k_{total} = k_1 + k_2$, $[H]_\infty = k_1[C_6D_5CH_3]_0/k_{total}$ and $BR_1 = k_1/k_{total}$. For the experiment of Fig. 1, $[C_6D_5CH_3]_0 = 1.66 \times 10^{12}$ molecules cm^{-3} giving $BR_1 = 1.46 \times 10^{12}/1.66 \times 10^{12} = 0.88$. With Eq. (6) analysis, the value determined for $k_{total} = 4400\ s^{-1}$, implying that $k_1 = 3872\ s^{-1}$. However, when the model in Table S1 is used to simulate the Fig. 1 experiment with k_1 and k_2 determined from first-order analysis, Eq. (6), it provides an excellent fit to the [H] profile rise but over-predicts the observed $[H]_\infty$ (shown by the dash–dot–dot line). This is due to additional H-atoms produced from $C_6D_5CH_2$ decay which is incorporated in the model and based on our benzyl study [22]. Consequently the best fit to this [H] profile, shown by the solid black line, is then obtained by making two adjustments to the model, lowering BR_1 by 0.10 and increasing k_{total} to 4500 s^{-1}. This implies that benzyl decomposition cannot be neglected in the experimental analysis and also that it is the branching ratio of H-atoms relative to the sum of H- and D-atoms from $C_6D_5CH_2$ that affects BR_1 and not the actual rate constants for benzyl decay. This is again evident in Fig. 2 where the [H] sensitivity analysis for the experiment in Fig. 1 is shown.

Figure 3 is another [H] profile from the thermal decomposition of $C_6D_5CH_3$ at 1713 K, and, for this experiment, the first-order analysis yields $k_{total} = 5800\ s^{-1}$ and $BR_1 = 1.41 \times 10^{12}/1.50 \times 10^{12} = 0.94$. Using the first-order analysis derived k_{total} and BR_1 in the model again fails to match the experimental $[H]_\infty$ (dash–dot–dot line in Fig. 3). Using these first-order values as a first guess with the model in Table S1, the best fit to the profile is obtained with $k_{total} = 6000\ s^{-1}$ and $BR_1 = 0.80$ (solid black line). Figure 3 also shows the results of a brute-force sensitivity analyses with the long dashed lines representing variations in the overall decomposition rate constant, k_{total}, by ±35% and with the dotted lines being variations of ±0.1 in the branching ratio giving H-atoms, $BR_1 = k_1/k_{total}$. For the entire dataset the modeled k_{total} is in excellent agreement, within ±10%, of the first-order analysis for k_{total}. However, the final modeled BR_1 is lowered by 0.10–0.15 from the derived values with first-order analysis. Thirty-three H-atom experiments have been performed between 1469 and 1859 K at densities, ≈1, 2, 3, and 6×10^{18} molecules cm^{-3}, and the conditions and the modeled results for k_{total}, k_1, k_2 and BR_1 are given in Table S2 (Supplementary data).

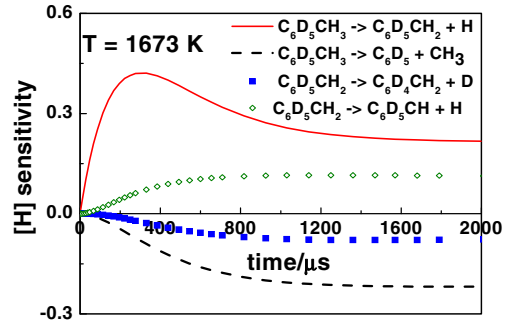

Fig. 2. H-atom sensitivity analysis for the profile in Fig. 1 using the Table S1 mechanism. The four most sensitive reactions are shown in the inset.

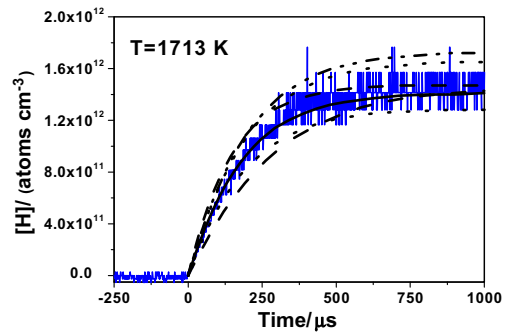

Fig. 3. [H] profile at 1713 K. Solid line – fit using the mechanism in Table S1 with $k_{total} = 6000\ s^{-1}$ and $BR_1 = 0.80$, dashed–dot–dot line – fit using the mechanism in Table S1 with $k_{total} = 5800\ s^{-1}$ and $BR_1 = 0.94$ from first-order analysis, dashed lines – $k_{total} \pm 35\%$ with $BR_1 = 0.80$, dotted lines – $BR_1 \pm 0.10$ with $k_{total} = 6000\ s^{-1}$. The conditions for the experiment are $P_1 = 15.92$ Torr, $M_s = 2.637$, $\rho_5 = 3.194 \times 10^{18}$ molecules cm^{-3}, $[C_6D_5CH_3]_0 = 1.500 \times 10^{12}$ molecules cm^{-3}.

D-atoms were also measured from the thermal decomposition of $C_6D_5CH_3$. Figure 4 is an example [D] profile for an experiment at 1705 K. The thermal decomposition rates from the present H-atom ARAS experiments were used as a first guess in Table S1 model to simulate the D-atom profile, and a reasonable fit to the observed [D] profile was obtained. However, getting a perfect fit to the profile is a complex optimization exercise involving at the least five strongly sensitive reactions in the model. This can be seen in Fig. 5 where the D-atom sensitivity analysis for the experiment in Fig. 4 is shown. The sensitivity of the D-atom profile is controlled by both $C_6D_5CH_3$ and all three $C_6D_5CH_2$ thermal decompositions over the entire time range of the experiments. Though the fit is

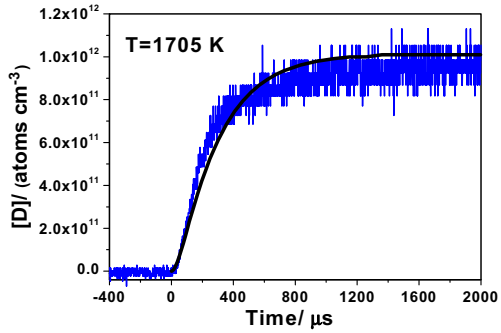

Fig. 4. [D] profile at 1705 K. Solid line – fit using the mechanism in Table S1 with $C_6D_5CH_3$ decomposition rate constants from the H-atom ARAS experiments. The conditions for the experiment are $P_1 = 10.93$ Torr, $M_s = 2.624$, $\rho_5 = 2.199 \times 10^{18}$ molecules cm^{-3}, $[C_6D_5CH_3]_0 = 1.682 \times 10^{12}$ molecules cm^{-3}.

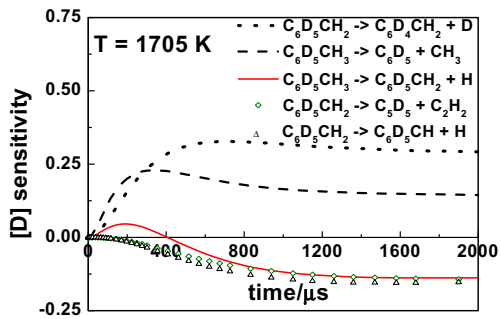

Fig. 5. D-atom sensitivity analysis for the profile in Fig. 4 using the Table S1 mechanism. The five most sensitive reactions are shown in the inset.

not perfect, the simulation depicted for the D-atom experiment is a cross-check for the measured rate constants for $C_6D_5CH_3$ decomposition in this work. The D-atom yields observed here are also consistent with the H-atom yields observed in the corresponding $C_6H_5CD_2$ studies [22] performed in this laboratory. Because of the obvious complexity and the strong interdependences, we have not attempted to fit/analyze the D-atom experiments presented here.

4. Discussion

There is an extensive database of high-T thermal decomposition rate measurements on toluene. However, since the major product, benzyl, dissociates faster than the parent, toluene, unambiguous experimental measurements of rate constants for toluene decomposition are extremely challenging. Previous high-T experimental measurements are complicated by multiple secondary reactions that include at the very least benzyl decomposition,

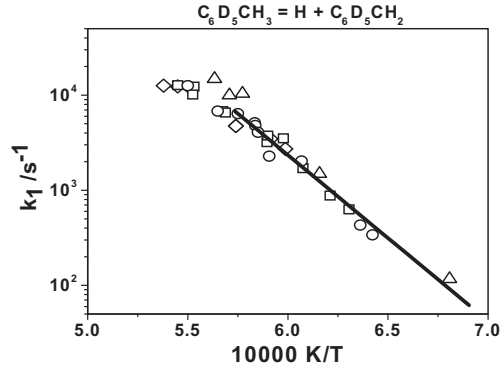

Fig. 6. Arrhenius plot for k_1. [◇] – $\rho_5 \sim 1 \times 10^{18}$ molecules cm^{-3}, [□] – $\rho_5 \sim 2 \times 10^{18}$ molecules cm^{-3}, [○] – $\rho_5 \sim 3 \times 10^{18}$ molecules cm^{-3}, [Δ] – $\rho_5 \sim 6 \times 10^{18}$ molecules cm^{-3}, [—] – Arrhenius fit, Eq. (7).

the fast recombination of H with benzyl and the reaction of H with the parent toluene. In the present work, interferences from secondary chemistry are minimized by observing H-atoms from the partially deuterated toluene-d_5 thereby suppressing the contributions from benzyl thermal decomposition. Furthermore, by using dilute amounts (<1 ppm) of the reactant, toluene-d_5, contributions from other secondary reactions are suppressed. The H-atom experiments presented here therefore provide the best estimates of *chemical isolation* rate constants for the primary dissociation channels (1) and (2). The experimental data from the present work have been used to obtain rate parameters for the two primary dissociation channels and for total toluene-d_5 decomposition.

An Arrhenius plot of the first-order rate constants for channel (1) from Table S2 is shown in Fig. 6. At $T < 1700$ K there is no discernible pressure dependence among the four data sets within experimental error. At higher-T the data at $\rho = 1$–3×10^{18} molecules cm^{-3} ($P \approx 0.2$–0.7 atm) show more fall-off than that with $\rho = 6 \times 10^{18}$ molecules cm^{-3} ($P \approx 1.5$ atm). The rate constants in the lower-T range can be represented to within ±35% in units of s^{-1} as,

$$k_1 = 6.91 \times 10^{13} \exp(-40180/T) \quad (1469\text{–}1714\ \text{K}) \tag{7}$$

Calculations with Eq. (7) give the solid line in Fig. 6. Similar pressure dependence is observed for channel (2). The lower-T results for k_2 give to within ±60%

$$k_2 = 5.99 \times 10^{15} \exp(-50060/T) \quad (1469\text{–}1714\ \text{K}) \tag{8}$$

and the total decomposition rate constants for toluene-d_5 (the sum of k_1 and k_2) give to within ±40%

$$k_{\text{Total}} = 2.21 \times 10^{14} \exp(-41760/T) \quad (1469\text{–}1714\,\text{K}) \tag{9}$$

where the units are s^{-1}.

Figure 7 shows the branching ratios (defined as k_1/k_{Total}) in the thermal decomposition of toluene-d_5. The present data, within experimental error, show little or no pressure dependence in the branching ratios. However, a small T-dependence might be indicated at $T < 1600$ K with a branching ratio ≈0.9, decreasing to ~0.75 at even higher-T. There are no previous thermal decomposition measurements on toluene-d_5. Theoretical calculations (S.J. Klippenstein, private communication, December 2009) suggest that the isotope effect is minor (<5%) over the range of pressures and temperatures in this study, and consequently this allows comparisons to the numerous protonated toluene studies. Since recent studies [19,37, and references therein] have contributed to a better understanding of the secondary chemistry in toluene decomposition, comparisons are made to only the two recent high-T studies [17,19]. The magnitudes as well as the temperature and pressure dependence of the branching ratios obtained in the present experiments are in excellent agreement with theoretical predictions for branching ratios in $C_6H_5CH_3$ by Klippenstein et al. [21]. Results from the experimental studies by Eng et al. [17] at 1.6 atm and Oehlschlaeger et al. [19] at 1.5 atm favor the phenyl channel (reaction (2)) more than indicated by the present experiments and the theoretical conclusions of Klippenstein et al. [21].

The total rate constants for $C_6D_5CH_3$ decomposition, $k_{\text{Total}} = k_1 + k_2$, are plotted in Fig. 8 where the recent experimental [17,19] and theoretical estimates [21] for k_{Total} are also summarized. The present rate constants are shown as symbols for the four different densities. The theoretical results in Table 1 of Klippenstein et al. [21] report incorrect values for k_0 and F_{cent} parameters (S.J. Klippenstein, private communication, December 2009), and consequently these have been corrected and are given in Table S3 (Supplementary data), for use by modelers. The present experimental results at high-T (>1700 K) are in the fall-off regime with the highest pressure data (1.5 atm) being in excellent agreement (within ±25%) with the 1.5 atm theoretical predictions of Klippenstein et al. [21]. The extent of fall-off observed in the present low-P experiments (0.2–0.7 atm) is also in excellent agreement with theory [21]. The 1.5 atm experimental k_{Total} results of Oehlschlaeger et al. [19] do not show fall-off over the entire T range (1400–1780 K) and are 40–70% higher than the theoretical k_∞ of Klippenstein et al. [21]. Above 1700 K, the k_{Total} results of Oehlschlaeger et al. [19] are close to three times higher than the present experimental results at 1.5 atm. Oehlschlaeger et al. [19] used UV-absorption of benzyl radicals at 266 nm as the diagnostic and therefore required fairly concentrated toluene mixtures (100–400 ppm). They have pointed out that their results are influenced by unavoidable interference from two secondary reactions, the thermal decomposition of benzyl and the reaction of H with toluene. Hence, they relied on their own measurements of these two rate constants for obtaining k_1 and k_2. We pointed out in our recent study of the thermal decomposition of benzyl [22] that a likely complication with these UV-absorption measurements is the formation of

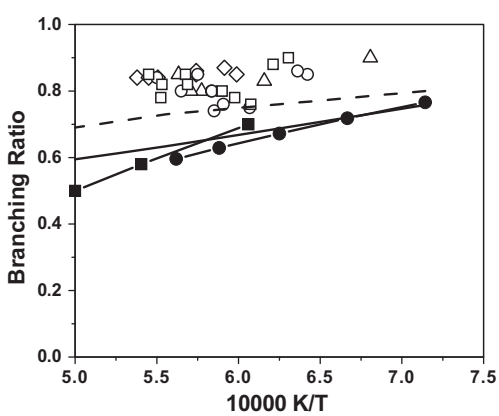

Fig. 7. Branching ratios in toluene and toluene-d_5. [◇] – $\rho_5 \sim 1 \times 10^{18}$ molecules cm^{-3}, [□] – $\rho_5 \sim 2 \times 10^{18}$ molecules cm^{-3}, [○] – $\rho_5 \sim 3 \times 10^{18}$ molecules cm^{-3}, [Δ] – $\rho_5 \sim 6 \times 10^{18}$ molecules cm^{-3}, [- - -] – $P = 1.5$ atm, Klippenstein et al. [21], [—] – $P = \infty$, Klippenstein et al. [21], [● — ●] – $P = 1.5$ atm, Oehlschlaeger et al. [19], [■ — ■] – $P = 1.6$ atm, Eng et al. [17].

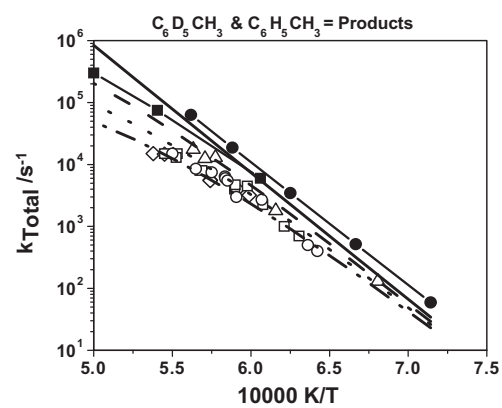

Fig. 8. Arrhenius plot for k_{Total}. [◇] – $\rho_5 \sim 1 \times 10^{18}$ molecules cm^{-3}, [□] – $\rho_5 \sim 2 \times 10^{18}$ molecules cm^{-3}, [○] – $\rho_5 \sim 3 \times 10^{18}$ molecules cm^{-3}, [Δ] – $\rho_5 \sim 6 \times 10^{18}$ molecules cm^{-3}, [– · –] – $P = 0.1$ atm, [···] – $P = 0.3$ atm, [- - -] – $P = 1.5$ atm, [—] – $P = \infty$, Dashed–dot, dotted, dashed and solid lines are predictions of Klippenstein et al. [21], [● — ●] – $P = 1.5$ atm, Oehlschlaeger et al. [19], [■ — ■] – $P = 1.6$ atm, Eng et al. [17].

multiple C_7H_7 isomers. If these exist then attributing absorption to only benzyl may be ambiguous, particularly if the spectra of these C_7H_7 isomers overlap that of benzyl. This could be a possible reason for the disagreement of the Oehlschlaeger at al. [19] results with the H-ARAS data from the present study, the H-ARAS data from Eng et al. [17], and the theoretical predictions of Klippenstein et al. [21]. The Eng et al. [17] predictions for k_{Total} at 1.6 atm are in good agreement with the theoretical k_∞ of Klippenstein et al. [21] at lower-T. However, at the higher-T range above 1700 K, their results are approximately 40–50% higher than the present 1.5 atm values. On the other hand at 0.1 atm, the Eng et al. [17] results for k_1 are in good agreement with the lower-P (0.2–0.3 atm) values from the present study. Similar dilute reactant mixtures as in the present study were used in these lower-P H-ARAS experiments [17], thereby suppressing secondary chemistry. Secondary reaction complications were probably more important in their higher-P experiments where 5–10 times more reactant was used.

The present study using deuterated toluene and ultra-sensitive H-atom ARAS as the diagnostic has permitted the best possible *chemical isolation* measurements of branching ratios and rate constants to be made in the two-channel thermal decomposition of toluene-d_5. Since the isotope effects are minimal (S.J. Klippenstein, private communication, December 2009), the present experimental results therefore are direct measurements of branching ratios and rate constants in toluene decomposition. The present results are in excellent agreement with the theoretical predictions of Klippenstein et al. [21] for both branching ratios and rate constants.

5. Conclusions

The thermal decomposition of toluene-d_5 ($C_6D_5CH_3$) has been studied at high temperatures with the reflected shock-tube technique using H- and D-atom ARAS. The experiments were performed at high-T (1469–1859 K) at nominal pressures ≈0.25–1.50 atm. The present study utilizing the ultra-sensitive H-atom ARAS technique has provided a direct measurement of the branching ratio in the two-channel high-temperature thermal decomposition of toluene-d_5. Since the isotope effects are minimal these experimental results therefore are direct measurements of branching ratios and rate constants in toluene decomposition. The present results are in excellent agreement with the recent theoretical predictions of Klippenstein et al. for the branching ratios and rate constants and lead us to conclude that the high-T thermal decomposition of toluene is now well-characterized.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.077.

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